

Laboratório — Sessão 4

Gestão Segura de Acessos Remotos SSH em Linux

Proteger o canal de gestão remota do servidor Ubuntu, eliminando a autenticação tradicional por password e migrando para autenticação criptográfica.

Ambiente Virtual

- TryHackMe — Linux Strength Training (gratuito): <https://tryhackme.com/room/linuxstrengthtraining>
- KillerCoda Ubuntu Playground: <https://killercoda.com/playgrounds/scenario/ubuntu>

Tarefas a Executar

Gerar um par de chaves Ed25519

```
root@ip-10-114-138-64:~# ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (/root/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_ed25519
Your public key has been saved in /root/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:MrpARft0FIadSsJCELI2f3bJM2Vbzzm6Hdf98+3xNt4 root@ip-10-114-138-64
The key's randomart image is:
+--[ED25519 256]--+
|*+ .oo.          |
|o.o.ooo.         |
|. + oo.. o .     |
|. o..o. + o o .  |
|.. o=*S. =       |
|. o+.oo . . o    |
|. . . . . oo     |
|. . . o ooB      |
|. . . . .+E      |
+-----[SHA256]-----+
```

Transferir a chave pública para o servidor alvo:

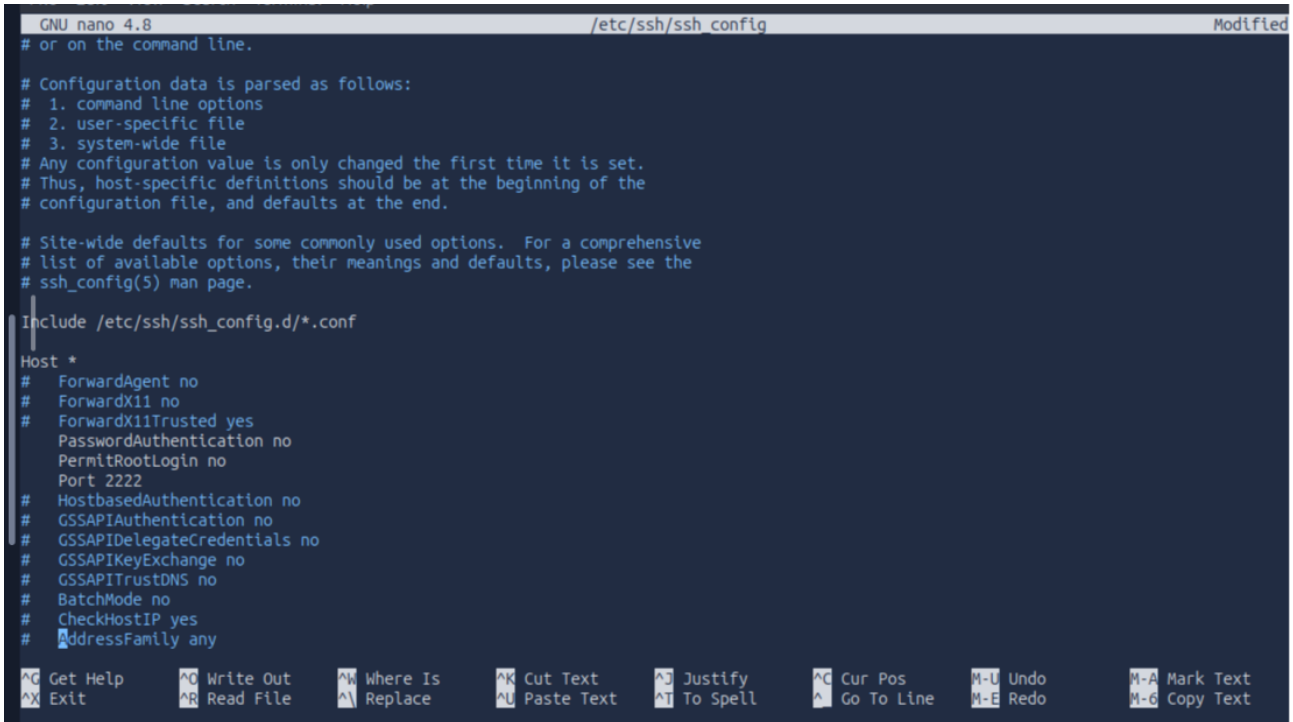
```
root@ip-10-114-138-64:~# ssh-copy-id fred@10.114.160.128
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_ed25519.pub"
The authenticity of host '10.114.160.128 (10.114.160.128)' can't be established.
ED25519 key fingerprint is SHA256:eescxeN6SI0ikAS07rSeLFrl7WY11AxbKKEZ1QLvMrw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that
are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is
to install the new keys
fred@10.114.160.128's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'fred@10.114.160.128'"
and check to make sure that only the key(s) you wanted were added.
```

Editar o ficheiro de configuração do daemon SSH com privilégios superutilizador:

Aplicar as seguintes alterações:



```
GNU nano 4.8 /etc/ssh/ssh_config Modified
# or on the command line.

# Configuration data is parsed as follows:
# 1. command line options
# 2. user-specific file
# 3. system-wide file
# Any configuration value is only changed the first time it is set.
# Thus, host-specific definitions should be at the beginning of the
# configuration file, and defaults at the end.

# Site-wide defaults for some commonly used options. For a comprehensive
# list of available options, their meanings and defaults, please see the
# ssh_config(5) man page.

Include /etc/ssh/ssh_config.d/*.conf

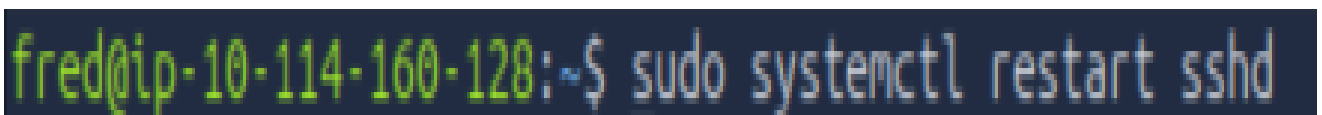
Host *
# ForwardAgent no
# ForwardX11 no
# ForwardX11Trusted yes
# PasswordAuthentication no
# PermitRootLogin no
# Port 2222
# HostbasedAuthentication no
# GSSAPIAuthentication no
# GSSAPIDelegateCredentials no
# GSSAPIKeyExchange no
# GSSAPITrustDNS no
# BatchMode no
# CheckHostIP yes
# AddressFamily any
```

Validar a sintaxe das alterações antes de reiniciar o serviço:



```
fred@ip-10-114-160-128:~$ sudo sshd -t
```

Reiniciar o serviço SSH:



```
fred@ip-10-114-160-128:~$ sudo systemctl restart sshd
```

1. Cópia das linhas modificadas do *sshd_config*

Ficheiro *sshd_config* inicial

```
GNU nano 4.8 /etc/ssh/ssh_config
# This is the ssh client system-wide configuration file. See
# ssh_config(5) for more information. This file provides defaults for
# users, and the values can be changed in per-user configuration files
# or on the command line.

# Configuration data is parsed as follows:
# 1. command line options
# 2. user-specific file
# 3. system-wide file
# Any configuration value is only changed the first time it is set.
# Thus, host-specific definitions should be at the beginning of the
# configuration file, and defaults at the end.

# Site-wide defaults for some commonly used options. For a comprehensive
# list of available options, their meanings and defaults, please see the
# ssh_config(5) man page.

Include /etc/ssh/ssh_config.d/*.conf

Host *
# ForwardAgent no
# ForwardX11 no
# ForwardX11Trusted yes
# PasswordAuthentication yes
# HostbasedAuthentication no
# GSSAPIAuthentication no
# GSSAPIDelegateCredentials no
# GSSAPIKeyExchange no
# GSSAPITrustDNS no
# BatchMode no
```

Read 52 lines

Get Help Write Out Where Is Cut Text Justify Cur Pos Undo Mark Text
Exit Read File Replace Paste Text To Spell Go To Line Redo Copy Text

Ficheiro *sshd_config* modificado

```
GNU nano 4.8 /etc/ssh/ssh_config Modified
# or on the command line.

# Configuration data is parsed as follows:
# 1. command line options
# 2. user-specific file
# 3. system-wide file
# Any configuration value is only changed the first time it is set.
# Thus, host-specific definitions should be at the beginning of the
# configuration file, and defaults at the end.

# Site-wide defaults for some commonly used options. For a comprehensive
# list of available options, their meanings and defaults, please see the
# ssh_config(5) man page.

Include /etc/ssh/ssh_config.d/*.conf

Host *
# ForwardAgent no
# ForwardX11 no
# ForwardX11Trusted yes
# PasswordAuthentication no
# PermitRootLogin no
# Port 2222
# HostbasedAuthentication no
# GSSAPIAuthentication no
# GSSAPIDelegateCredentials no
# GSSAPIKeyExchange no
# GSSAPITrustDNS no
# BatchMode no
# CheckHostIP yes
# AddressFamily any
```

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2. Evidência de login bem-sucedido via chave criptográfica (output do terminal)

```
root@ip-10-114-138-64:~# ssh fred@10.114.160.128
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun 26 Jul 19:35:19 UTC 2026

System load:  0.04               Processes:    113
Usage of /:   61.6% of 8.76GB    Users logged in: 0
Memory usage: 8%                IPv4 address for ens5: 10.114.160.128
Swap usage:   0%

 * Ubuntu 20.04 LTS Focal Fossa has reached its end of standard support on 31 Ma
 |
 | For more details see:
 | https://ubuntu.com/20-04

Expanded Security Maintenance for Infrastructure is not enabled.

8 updates can be applied immediately.
2 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

129 additional security updates can be applied with ESM Infra.
Learn more about enabling ESM Infra service for Ubuntu 20.04 at
https://ubuntu.com/20-04

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Your Hardware Enablement Stack (HWE) is supported until April 2025.

Last login: Tue Apr 20 10:14:19 2021
```